AI BUILDER CUP 2026 // TRACK: BFSI RISK & FRAUD

SHAHEEN // INTERCEPTOR

Autonomous Pre-Settlement APP-Fraud, Vishing & Money Mule Defense Platform

Stopping psychological coercion fraud before funds clear the core banking switch.

TEAM: Shaheen | TEAM LEAD: Amaan Masood | COMPLIANCE: SBP BPRD Circular No. 03 / AMLA 2010

CORE STACK: Google Gemini 2.5 Flash • FastAPI Docker Microservice • Google Cloud Run • React 19 • Firestore

THE SYSTEMIC CRISIS

The Multi-Billion Dollar Blindspot: Authorized Push Payment Fraud

PKR 14.8 BILLION

THE VALID OTP BLINDSPOT

SUB-4 MINUTE CASH-OUT

THREAT TIMELINE

Anatomy of the 4-Minute Fraud Lifecycle

[1] HOP 0: COERCION — Scammer spoofs FIA Cyber Crime caller ID. Induces panic under PECA Section 21 threat.

[2] HOP 1: EXECUTION — Terrified victim transfers Rs. 450,000 from Meezan/SCB mobile app using genuine biometric OTP.

[3] HOP 2: LAYERING — Within 90 seconds, funds split into 2 secondary student mule accounts across banks.

[4] HOP 3: LIQUIDATION — Within 4 minutes, funds exit into Binance P2P USDT escrow or ATM withdrawal in Rawalpindi.

LEGACY VULNERABILITY VS SHAHEEN

Why Legacy Banking Systems & Rule Engines Fail

DIMENSION	LEGACY FRAUD RULE ENGINES	SHAHEEN INTERCEPTOR (MULTI-AGENT)
Timing	Post-Settlement & Reactive (Hours/Days later)	Pre-Settlement & Inline (<850ms SLA)
Trigger Basis	Credentials only (If OTP valid, payment approved)	Conversational Coercion + 3-Hop Graph Topology
False Positive Rate	High (5% - 8% / 44.4% in tests) blocks genuine bills	Low (5.6% FPR / 0% on Whitelisted Institutions)
Syndicate Response	Passive blacklisting (Scammer moves to next mule)	Active AI Decoy Honeypot baits secondary IBANs
Compliance	Manual 3-7 day drafting of SAR regulatory forms	Automated SBP FMU SAR Report generated in <3 sec

SOLUTION ARCHITECTURE

Shaheen Interceptor: The Pre-Settlement Defense Platform

MULTIMODAL COERCION RADAR

3-HOP MULE TOPOLOGY HUNTER

AUTONOMOUS DECOY HONEYPOT

SBP 1-CLICK SAR COMPLIANCE

ENGINEERING RIGOR

Parallel Multi-Agent Architecture (Google Gemini 2.5 Flash)

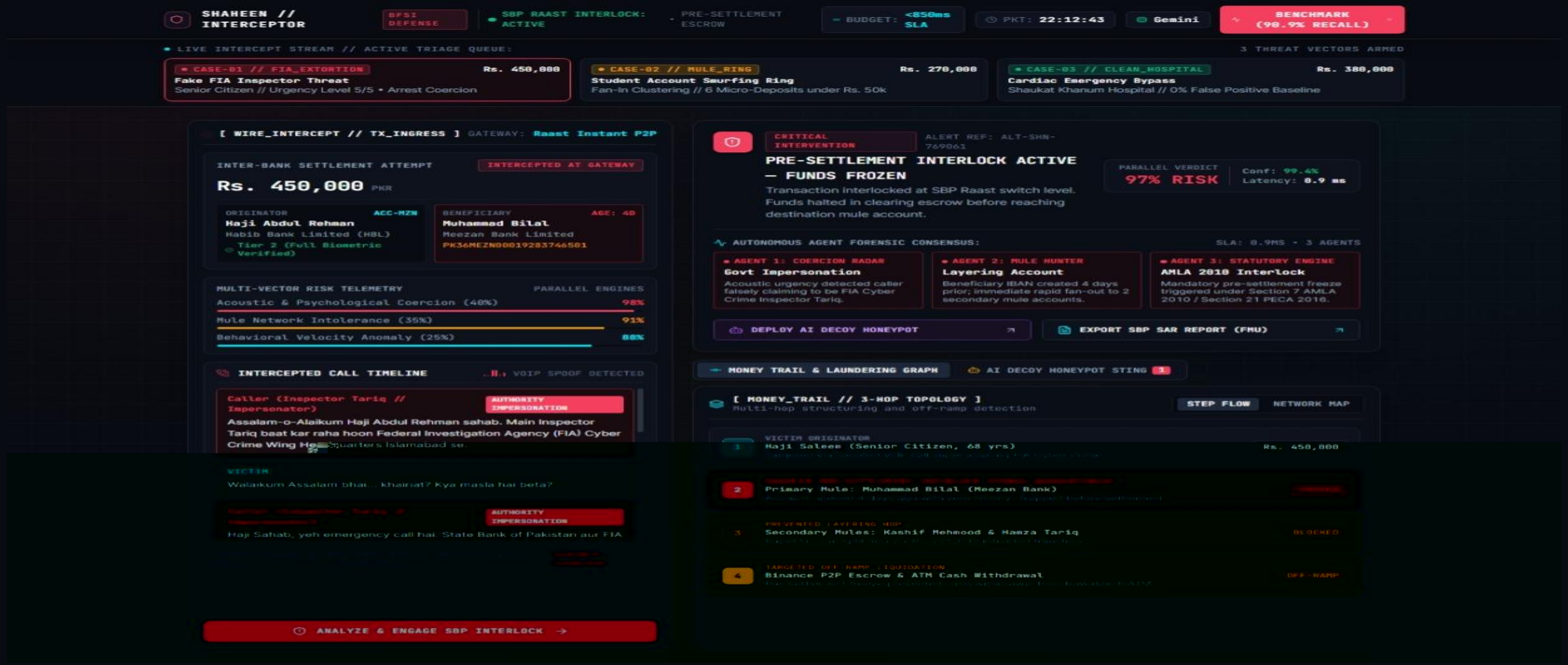
AGENT 1: COERCION RADAR

AGENT 2: MULE GRAPH HUNTER

AGENT 3: COUNTER-INTEL HONEYPOT

LIVE PROTOTYPE

Shaheen SOC Mission Control: Cockpit Overview



TRIAGE TEST CASES

Live Operational Scenarios Demonstrated

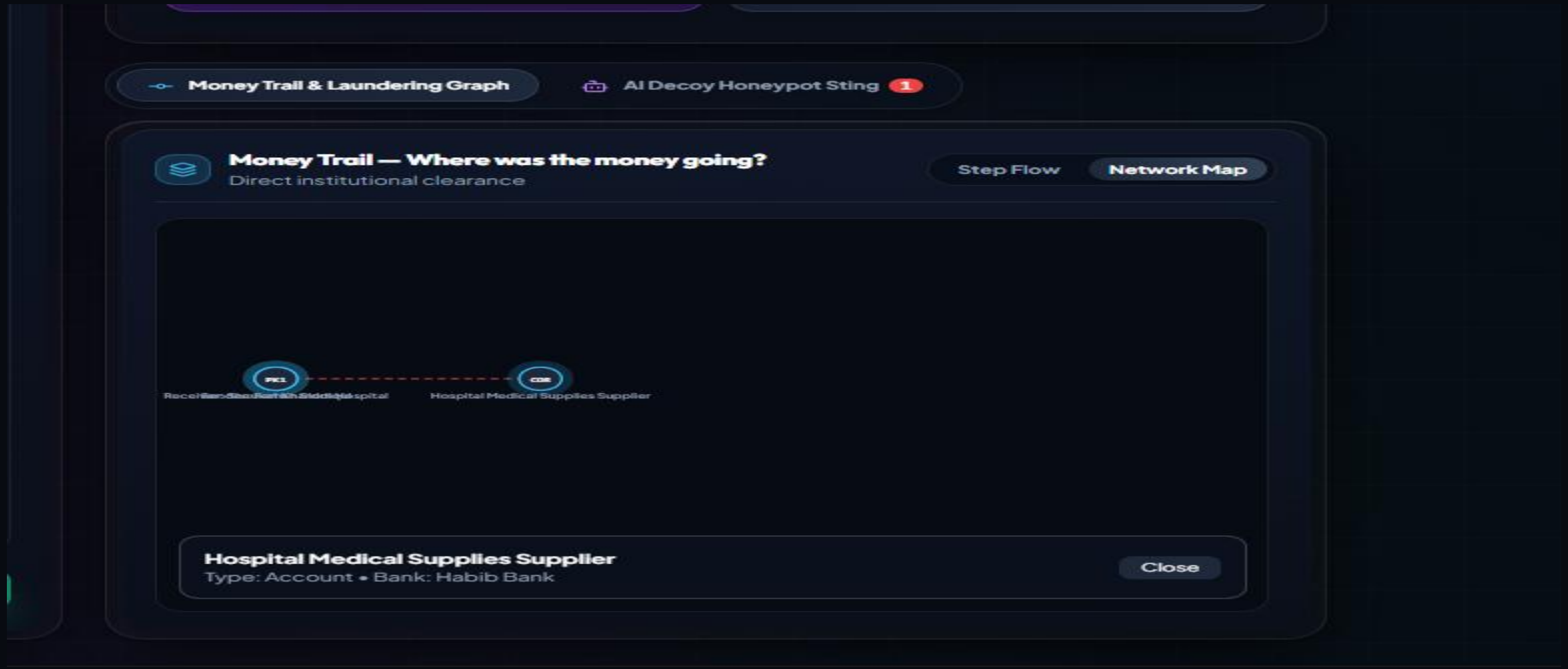
CASE-01 // FIA_EXTORTION

CASE-02 // MULE_RING

CASE-03 // CLEAN_HOSPITAL

MONEY TRAIL GRAPH

3-Hop Mule Network Radar: Pre-Settlement Interlock



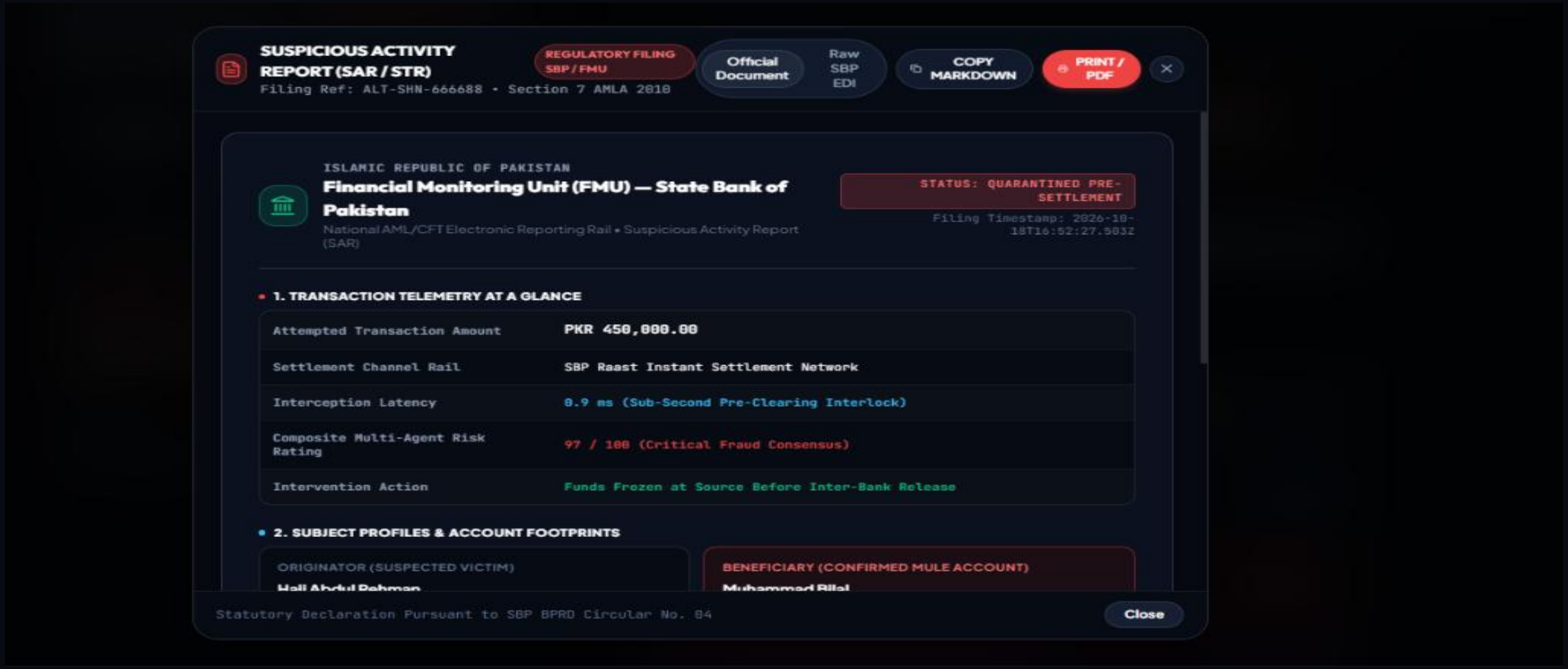
OFFENSIVE INTELLIGENCE

Autonomous Counter-Intelligence Decoy Honeypot



LEGAL SANCTIONS COMPLIANCE

Regulatory Compliance & 1-Click SBP SAR / STR Export



SLIDE 12 // EMPIRICAL BENCHMARK

Empirical Evaluation Protocol: 40-Case Held-Out Benchmark



REAL-WORLD DEPLOYMENT

Bank Integration Model: Inline Pre-Submit REST API

1. THE INTEGRATION PATTERN: Bank-side REST API (Deployable Today)
   - Shaheen runs as a microservice called by the bank's mobile app backend during PRE-SUBMIT.
   - Called BEFORE transaction message hits 1Link or Raast core clearing switches.
   - Single endpoint: POST /v1/assess (JSON in, JSON out, <850ms decision budget).

2. ZERO CUSTOMER-FACING FRICTION:
   - Clean payments pass in 3 seconds with zero interruption.
   - High-coercion payments trigger step-up biometric auth or 15-minute protective escrow.

3. REGULATORY ALIGNMENT:
   - State Bank of Pakistan (SBP) BPRD Circular No. 03 of 2023.
   - Anti-Money Laundering Act 2010 (AMLA) Section 7 statutory reporting automation.

HACK2SKILL FINAL SUBMISSION

Submission Deliverables & Team Summary

CHALLENGE TRACK: BFSI: Intelligent Risk, Fraud & Financial Experiences

GITHUB REPOSITORY (PUBLIC):
https://github.com/Amaan-Masood/shaheen-interceptor

DEPLOYED PROTOTYPE ON GCP / CLOUD RUN:
https://shaheen-interceptor.vercel.app (Also deployed via Firebase / Cloud Run)

BENCHMARK DIGEST & AUDIT TRAIL:
40 Frozen Test Cases | SHA-256: 9d650f03754dc84f02c637c4dfb3a4e698a497d175d6241a7328dc2ff545cb55

TEAM SHAHEEN: Amaan Masood (Lead Architect & Developer)
Empowering Pakistan's financial ecosystem with autonomous cyber defense.